Fiscal Year:	FY 2004	**Task Last Updated:**	FY 02/24/2006
PI Name:	Dietrich, Daniel L Ph.D.		
Project Title:	Diffusion Flame Suppression in Microgravity using Spray Agents		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44135	**Congressional District:**	10
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	01/01/2003	**End Date:**	09/30/2006
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Takahashi, Fumiaki (NASA Glenn Research Center)		
Grant/Contract No.:	In-House		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Despite their relevance, very few studies exist regarding the effectiveness of spray agents in simple geometries where a fundamental understanding of their action can be achieved. This is especially true for microgravity applications. Therefore, this proposal will examine the effectiveness of sprayed agents in extinguishing a model diffusion flame in microgravity. The goal is to improve our ability to extinguish fires and develop effective fire-fighting strategies aboard spacecraft. The study will involve both normal and microgravity experiments and detailed numerical modeling of the flame. The experiments will be gaseous counter-flow diffusion flames, and condensed phase (solid and liquid) stagnation point diffusion flames. We will use a variety of diagnostics on the flames including conventional photography, species specific chemi-luminescent imaging, particle image velocimetry and thin filament pyrometry. All of these diagnostics are currently available in the investigators laboratories. In addition, we propose to purchase a single component Phase		

Task Description:	Doppler Particle Analyzer to perform a detailed characterization of the flame. The experiments will be conducted over a wide range of relevant experiment parameters including oxygen mole fractions, flame strain rates, ambient pressures and fuel types. The fuel sprays will be from commercially available atomizers and will further consist of both monodisperse and polydisperse sprays. The monodisperse sprays, while not practical are much more amenable to detailed numerical modeling. The numerical modeling effort is based upon an existing numerical model under development by V. Katta of Innovative Scientific Solutions, Inc. It is a two-dimensional transient code that includes a detailed treatment of the fluid mechanics, and chemical kinetics. The treatment of the spray will be incorporated into the model as a part of the proposed program.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	For more information on this task, please contact the Task Book Help Desk at taskbook@nasaprs.com.
Bibliography Type:	Description: (Last Updated: 02/27/2004)